

1. Various Types of AI

AI is generally categorized based on two main criteria: **Capabilities** and **Functionalities**.

- **Based on Capabilities:**
 - **Narrow AI (Weak AI):** Specialized in one task (e.g., Siri, Alexa, or recommendation systems).
 - **General AI (Strong AI):** Theoretical AI that can perform any intellectual task a human can.
 - **Super AI:** A hypothetical stage where AI surpasses human intelligence across all fields.
- **Based on Functionalities:**
 - **Reactive Machines:** No memory; they only respond to current scenarios (e.g., Deep Blue chess computer).
 - **Limited Memory:** Can use past data to make better decisions (e.g., Self-driving cars).
 - **Theory of Mind:** Theoretical AI that understands human emotions and beliefs.
 - **Self-Awareness:** The ultimate stage where AI has its own consciousness and self-awareness.

2. Differentiate between Supervised and Unsupervised Learning

Feature	Supervised Learning	Unsupervised Learning
1. Definition	A type of machine learning where the model is trained on a labeled dataset	A type of machine learning where the model identifies hidden

	(input-output pairs) to predict future outcomes.	patterns or structures within unlabeled data.
2. Goal	To map an input (\$x\$) to an output (\$y\$) based on provided examples.	To find the underlying structure or distribution in the data.
3. Supervision	Operates under the guidance of a "teacher" or labeled data.	Operates on its own to discover information.
4. Types	Categorized into Regression and Classification.	Categorized into Clustering and Association.
5. Accuracy	Highly accurate as it uses verified labels for training.	Less accurate; results are subjective and depend on pattern discovery.
6. Examples	Spam detection (Gmail), House price prediction.	Customer segmentation (Netflix), Image compression.

3. Informed vs. Uninformed Search Algorithms

Feature	Uninformed Search (Blind Search)	Informed Search (Heuristic Search)
1. Definition	A search strategy that explores the search space without any domain knowledge or information about the distance to the goal.	A search strategy that uses heuristic functions (additional information) to estimate the cost/distance to the goal.
2. Knowledge	Knows only the start state, the goal state, and valid operators.	Uses domain-specific knowledge to guide the search towards the goal efficiently.
3. Performance	Generally slower and consumes more memory (explores all directions).	Faster and more memory-efficient as it focuses on the most promising paths.
4. Heuristic ($h(n)$)	Does not use a heuristic function.	Uses a heuristic function $h(n)$ to rank nodes.
5. Complexity	High time and space complexity.	Lower complexity due to targeted exploration.
6. Examples	Breadth-First Search (BFS),	A* Search, Greedy Best-First

	Depth-First Search (DFS).	Search.
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4. Application Areas of AI

AI is revolutionizing multiple industries:

- **Healthcare:** Medical imaging analysis, drug discovery, and robotic surgery.
- **Finance:** Fraud detection, algorithmic trading, and credit scoring.
- **E-commerce:** Personalized recommendations (Amazon/Netflix) and chatbots.
- **Automotive:** Self-driving cars and traffic management systems.
- **Education:** Personalized learning paths and automated grading.
- **Gaming:** Creating intelligent Non-Player Characters (NPCs).

5. PEAS Description for a House-Cleaning Robot

PEAS stands for **P**erformance, **E**nvironment, **A**ctuators, and **S**ensors. It is used to define the task environment of an intelligent agent.

For a House-Cleaning Robot:

- **P (Performance Measure):** Cleanliness of the floor, time taken, battery efficiency, safety (not hitting pets/objects).
 - **E (Environment):** Rooms, hallways, carpets, furniture, stairs, and inhabitants (humans/pets).
 - **A (Actuators):** Wheels or tracks for movement, brushes for sweeping, vacuum motor, water spray for mopping.
 - **S (Sensors):** Camera, Lidar/Ultrasonic sensors (for obstacles), Cliff sensors (to avoid stairs), Bump sensors, Dust sensors.
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6. Problems Robots Can Solve

Robots are primarily used to solve problems categorized as the "4 Ds":

- **Dirty:** Tasks like sewage cleaning, house cleaning, or waste management.
 - **Dull:** Repetitive assembly line tasks in manufacturing (e.g., picking and placing).
 - **Dangerous:** Defusing bombs, deep-sea exploration, or space missions.
 - **Difficult:** High-precision tasks like microsurgery or heavy lifting in logistics.
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7. Components of an AI Program

An AI program typically consists of several core technical components:

1. **Input/Perception:** Gathering data through sensors or digital inputs (Data Collection).
2. **Knowledge Base:** A repository of facts, rules, and data that the AI uses to understand the domain.
3. **Inference Engine:** The "brain" that applies logical rules to the knowledge base to derive new information or make decisions.
4. **Learning Component:** Algorithms that allow the system to improve its performance over time based on experience.
5. **Output/Actuation:** The final action taken, whether it's a digital response (chatbot) or physical movement (robot).

In AI, the **PEAS description** is a framework used to specify the characteristics of an intelligent agent and its task environment. It is the first step in designing any rational agent.

8. Describe PEAS description . Illustrate the PEAS description for Vacuum Cleaner Agent.

Breakdown of PEAS

- **Performance Measure:** The criteria that determine how successful the agent is. It defines what "success" looks like.
 - **Environment :** The external surroundings or world in which the agent operates.
 - **Actuators :** The "parts" that allow the agent to act upon the environment or perform tasks.
 - **Sensors:** The "inputs" that allow the agent to perceive its environment and gather data.
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2. PEAS Description for a Vacuum Cleaner Agent

For an automated vacuum cleaner (like a Roomba), the PEAS description identifies its goals, its working area, and how it interacts with its surroundings.

Component	Description for Vacuum Cleaner Agent
Performance	Cleanliness (percentage of dirt removed), time taken, battery consumption, and safety (not falling or breaking items).
Environment	Wooden floors, carpets, obstacles (furniture, shoes), stairs, pets, and walls.
Actuators	Motorized wheels for movement, rotating brushes for sweeping, vacuum motor for suction, and speakers/lights for status.
Sensors	Camera/LiDAR for mapping, Dirt sensors to detect debris, Bump sensors for obstacles, and Cliff sensors to avoid stairs.

9. Classify the types of chaining.

Illustrate forward chaining and backward chaining algorithms with suitable examples.

10.Explain the PEAS description for Automated taxi driver.

- **Performance (P):** The goals or "success criteria"—**Safety** (avoiding crashes), **Speed** (fastest route), **Legality** (obeying traffic laws), and **Passenger Comfort** (smooth ride).
- **Environment (E):** The external surroundings—**Roads** (highways/streets), **Traffic** (cars/buses), **Pedestrians**, **Weather** (rain/fog), and **Street Signs**.
- **Actuators (A):** The parts used to act/move—**Steering Wheel**, **Accelerator**, **Brakes**, **Signal Lights**, and **Horn**.
- **Sensors (S):** The parts used to perceive—**Cameras** (traffic lights/lanes), **LiDAR/Radar**

(distance to objects), **GPS** (location), and **Speedometer**.

Knowledge-based agents are intelligent agents that reason about their world using a formal internal representation of knowledge. According to **GeeksforGeeks** and standard AI theory (Russell & Norvig), there are three distinct levels at which we can describe these agents.

11. Describe levels of knowledge based agents.

The Three Levels of Knowledge-Based Agents

Level	Description	Key Focus
1. Knowledge Level	The highest and most abstract level. It describes what the agent knows and what its goals are, regardless of how the knowledge is implemented.	Goals and Facts
2. Logical Level	The level where knowledge is encoded into formal sentences or logic (such as First-Order Logic). It defines how the facts are structured.	Representation
3. Implementation Level	The lowest level, focusing on the actual physical representation or data structures (like linked lists or arrays) inside the computer system.	Data Structures

Detailed Explanation

1. Knowledge Level (or Linguistic Level)

At this level, we specify the agent's behavior by describing what it knows. We don't care about the computer code; we only care about the information.

- **Example:** "The agent knows that if it is raining, the grass will be wet, and its goal is to keep the floor dry."

2. Logical Level (or Functional Level)

This level bridges the gap between abstract knowledge and computer code. Here, the knowledge is translated into logical notations. The agent uses an **Inference Engine** to manipulate these logical sentences to derive new information.

- **Example:** $\text{IsRaining}(x) \rightarrow \text{GrassIsWet}(x)$

3. Implementation Level (or Physical Level)

This is the architecture where the knowledge actually "lives" in the machine. It involves the specific programming language (Python, LISP, Prolog) and the data structures (symbols, hashes, or pointers) used to store the Knowledge Base (KB).

- **Example:** Using a Python dictionary or a SQL database to store facts and rules.
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Comparison of Forward and Backward Chaining

Feature	Forward Chaining	Backward Chaining
Approach	Data-driven	Goal-driven
Starting Point	Known facts	Specific goals
Efficiency	Efficient for exploring all possible inferences	Efficient for achieving specific goals
Memory Usage	Can be memory intensive due to storing intermediate facts	Typically requires less memory
Implementation Complexity	Simple to implement	More complex to implement due to recursive nature
Suitability	Suitable for dynamic environments with continuously arriving data	Suitable for diagnostic systems and interactive applications
Handling Large Rule Sets	May become slow with large rule sets	Can be complex to manage with large rule sets

Feature	Propositional Logic	First-Order Logic
Basic Unit	Propositions	Predicates, constants, variables
Expressiveness	Limited to true/false statements	Expressive, can represent relationships and properties
Quantifiers	None	Universal ($\forall$) and Existential ($\exists$)
Syntax	Combines propositions using logical connectives	Uses predicates and quantifiers
Semantics	Truth tables	Interpretation over a domain
Use Cases	Simple problems (e.g., circuit design, rule-based systems)	Complex problems (e.g., AI reasoning, ontology modeling)
Example	$P \rightarrow Q$	$\forall x \exists y (Likes(x, y))$

1. Write a short note on Goal based agent with a diagram

2. Define pruning? Explain alpha and beta pruning algorithm

3. Differentiate between first-order logic and propositional logic.

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#SEE NOTEBOOK

4. Define Belief Network. Describe the steps of constructing a belief network with an example. What types of inferences can be drawn from that?

Definition

A Belief Network (Bayesian Network) is a Directed Acyclic Graph (DAG) that represents variables as nodes and their conditional dependencies as directed edges. It uses Conditional Probability Tables (CPTs) to quantify these relationships.

Steps to Construct the Belief Network (from your example)

1. Identify the Variables:

In this example, the variables are Humid (H), Cloud (C), Rain (R), Wet Grass (W), and School Off (O).

2. Determine Causal Relationships:

- Humidity and Clouds are independent factors that influence Rain.
- Rain directly influences whether the Grass is Wet and whether School is Off.

3. Draw the Network Structure (Nodes and Edges):

- **Root Nodes:** Create nodes for Humid and Cloud (no incoming arrows).
- **Intermediate Node:** Create a Rain node with arrows pointing from Humid and Cloud to Rain.
- **Leaf Nodes:** Create Wet Grass and School Off nodes with arrows pointing from Rain to each.

4. Quantify with Conditional Probability Tables (CPTs):

Assign the probability values for each scenario:

- **Independent Probabilities:** $P(H) = 0.2$ and $P(C) = 0.6$.
- **Joint Probabilities (Rain):** Fill the CPT for Rain based on all combinations of H and C (e.g., $P(R | H, C) = 0.9$).
- **Dependent Probabilities:** Define probabilities for the effects (e.g., $P(W|R) = 0.8$, meaning if it rains, there is an 80% chance the grass is wet).

Types of Inferences from this Example

1. **Causal Inference (Top-Down):**
 - **Scenario:** It is Humid and Cloudy. What is the probability that it will Rain?
 - **Direction:** From parents (H, C) to child (R).
2. **Diagnostic Inference (Bottom-Up):**
 - **Scenario:** The grass is wet. What is the probability that it actually Rained?
 - **Direction:** From effect (W) back to cause (R).
3. **Intercausal Inference (Explaining Away):**
 - **Scenario:** It is Raining. If we discover it is very Cloudy, the probability that high Humidity was the cause "decreases" because the clouds explain the rain.
 - **Direction:** Between two causes (H and C) of the same effect (R).
4. **Mixed Inference:**
 - **Scenario:** Knowing the grass is wet (Diagnostic) and it is not cloudy (Causal), what is the probability of it being Humid?

#NOTEBOOK

5. Discuss A* Algorithm with an example.

1. Definition

A* is a best-first search algorithm that finds the shortest path from a given starting node to a target goal node. It is widely used in maps, GPS systems, and video game character movement.

2. The Core Formula

The algorithm decides which path to take by calculating a "cost" for each node using the following equation:

$$f(n) = g(n) + h(n)$$

- **$g(n)$** : The actual cost of the path from the **start node** to the current node n .
- **$h(n)$** : The **heuristic value**, which is an estimated cost from node n to the **goal**.
- **$f(n)$** : The total estimated cost of the lowest-cost solution through node n .

3. How A* Works (Step-by-Step)

1. **Initialize**: Place the starting node in an **Open List** (nodes to be explored).
2. **Select**: Pick the node with the lowest $f(n)$ value from the Open List.
3. **Expand**: Move this node to the **Closed List** (already explored). Check its neighbors.
4. **Update**: For each neighbor, calculate its g , h , and f values. If a neighbor is already in the Open List with a higher cost, update it with the lower cost.
5. **Repeat**: Continue this process until the goal node is reached or the Open List is empty (meaning no path exists).

4. Key Properties (Important for Exams)

- **Completeness**: A* is **complete**, meaning it will always find a path if one exists.
- **Optimality**: A* is **optimal** (finds the shortest path) if the heuristic $h(n)$ is **admissible**.
 - *Admissible* means the heuristic never overestimates the actual cost to reach the goal.
- **Efficiency**: It is more efficient than Dijkstra's Algorithm because it uses the heuristic to ignore paths that head in the wrong direction.

5. Heuristic Functions

The choice of $h(n)$ depends on the problem:

- **Manhattan Distance**: Used when movement is restricted to 4 directions (up, down, left, right).
 - **Euclidean Distance**: The straight-line "as the crow flies" distance, used when movement can be in any angle.
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6.Explain the concept of hill climbing algorithm & simulated annealing algorithm.

Local Search Algorithms: AI Notes

1. Hill Climbing Algorithm

Definition: A greedy local search algorithm that continuously moves in the direction of increasing value (upland) to find the peak of a mountain or the best solution. It terminates when it reaches a state where no neighbor has a higher value.

Key Concepts

- **Initial State:** Starts with an arbitrary solution.
- **Neighbor Evaluation:** Checks the immediate neighbors of the current state.
- **Greedy Choice:** Moves only if the neighbor is better than the current state.
- **Termination:** Stops at a "Local Optimum" where no further improvement is possible.

The Search Landscape (Common Obstacles)

- **Local Maxima:** A peak that is higher than its neighbors but lower than the global maximum.
- **Plateau:** A flat area where all neighbors have the same value, making it difficult to find a direction.
- **Ridges:** A sequence of local maxima where the surface slopes in all directions except the top of the ridge.

Variants

1. **Simple Hill Climbing:** Moves to the *first* neighbor that is better than the current state.
2. **Steepest-Ascent:** Evaluates *all* neighbors and chooses the one with the highest improvement.
3. **Stochastic Hill Climbing:** Chooses at random from among the better neighbors.

2. Simulated Annealing Algorithm

2. Simulated Annealing Algorithm

Definition: A probabilistic optimization algorithm inspired by the physical process of **annealing in metallurgy** (heating and slow cooling of metals). It avoids getting stuck in local optima by occasionally accepting "worse" moves.

The Annealing Concept

- **High Temperature (T):** At the start, the algorithm "explores" the landscape by accepting worse moves with high probability.
- **Cooling Schedule:** As T gradually decreases, the algorithm becomes more "greedy," eventually settling into a global optimum.

Mathematical Mechanism

1. **Energy Difference (ΔE):**

$$\Delta E = Cost(S_{neighbor}) - Cost(S_{current})$$

2. **Acceptance Probability (P):** If a move is worse ($\Delta E > 0$), it is accepted only if a random number between 0 and 1 is less than:

$$P = e^{-\frac{\Delta E}{T}}$$

3. **Cooling Function:** The temperature is reduced in each iteration using a constant α :

$$T_{new} = \alpha \times T_{old} \quad (\text{where } 0 < \alpha < 1)$$

Comparison Table: Hill Climbing vs. Simulated Annealing

Feature	Hill Climbing	Simulated Annealing
Strategy	Purely Greedy	Probabilistic / Metaheuristic

Move Choice	Only moves "upward"	Can move "downward" to explore
Local Maxima	Often gets stuck	Designed to escape local maxima
Parameters	None (Simple)	Requires T and Cooling Rate (α)
Outcome	Local Optimum	Likely to find Global Optimum
Analogy	Climbing in the fog	Metal cooling to a stable state

7. Write a short note on Statistical learning.

Definition

Statistical Learning is a branch of machine learning that focuses on understanding and modeling the relationship between input variables (features) and output variables (target) using statistical methods.

Key Idea

Statistical learning aims to learn a function of the form:

$$Y = f(X) + \varepsilon$$

Where:

- X = Input variables
- Y = Output variable

- ε = Random error

Goals:

- Prediction (accurate estimation of output)
 - Inference (understanding relationship between variables)
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Types of Statistical Learning

1. Supervised Learning
 - Uses labeled data
 - Examples: Regression, Classification
 2. Unsupervised Learning
 - Uses unlabeled data
 - Examples: Clustering, Dimensionality Reduction
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Common Techniques

- Linear Regression
 - Logistic Regression
 - Decision Trees
 - k-Nearest Neighbors (KNN)
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Applications

- Spam detection
 - Medical diagnosis
 - Stock market analysis
 - Recommendation systems
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Advantages

- Strong mathematical foundation
- Helps in interpreting data
- Useful for prediction and analysis

Limitations

- Assumes certain data distributions
- Sensitive to noise and outliers

Conclusion

Statistical learning combines statistics and machine learning to build models that can predict outcomes and extract meaningful insights from data.

1. List out foundation of AI and describe any two in detail.

The Foundations of AI

Discipline	Contribution to AI
Philosophy	Logic, methods of reasoning, and the mind-body problem.
Mathematics	Formal logic, probability, computation, and algorithms.
Economics	Decision theory, game theory, and maximizing utility.
Neuroscience	Understanding how the brain processes information.
Psychology	Studying human behavior, perception, and cognition.
Computer Engineering	Providing the hardware and software tools to run AI.
Control Theory	Designing systems that act optimally based on feedback.
Linguistics	Understanding language structure and natural

	language processing.
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Detailed Description of Two Foundations

1. Philosophy

Philosophy is the oldest foundation of AI. It addresses the fundamental questions of how the mind works and whether a machine can truly be "intelligent."

- **Rationalism vs. Empiricism:** Philosophers debated whether knowledge is innate or gained through experience. This influenced the development of **Knowledge-Based Systems** (pre-programmed rules) versus **Machine Learning** (learning from data).
- **Logic:** Philosophers like Aristotle and later George Boole developed formal systems of logic that allow us to represent facts and derive new information, which is the basis for symbolic AI.
- **The Mind-Body Problem:** This explores whether the mind is purely physical. If it is, then it should be possible to replicate it in a physical machine.

2. Mathematics

Mathematics provides the formal tools and language required to design and analyze AI algorithms. Without math, AI would only be a set of abstract ideas without a way to execute them.

- **Logic:** Mathematical logic provides the framework for AI to perform "reasoning."
 - **Computation:** Alan Turing's work on "computability" defined what a machine can and cannot do, establishing the limits of AI.
 - **Probability:** Most AI today (like ChatGPT or self-driving cars) deals with uncertainty. Probability theory allows AI to make the "best guess" when information is incomplete or noisy.
 - **Optimization:** Calculus and linear algebra are used to "train" models (like Neural Networks) by minimizing errors and finding the most accurate mathematical patterns in data.
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2.Explain various properties of the task environment with suitable examples.

/Explain the different types of environments of AI agents.

Definition: A task environment represents the "problem space" where an Artificial Intelligence agent operates. It includes everything external to the agent that influences its perceptions and actions. The nature of these environments is categorized by several key properties that determine the complexity of the agent's design.

1. Observability

- **Fully Observable:** The agent's sensors provide a complete state of the environment at each point in time. There is no "hidden" information.
 - *Example: Chess* (the entire board is visible).
- **Partially Observable:** The agent has incomplete or noisy data due to inaccurate sensors or missing information.
 - *Example: Poker* (opponent's cards are hidden).

2. Certainty (Nature)

- **Deterministic:** The next state of the environment is completely determined by the current state and the action executed by the agent.
 - *Example: Sudoku* (filling a number leads to a fixed result).
- **Stochastic:** The environment involves randomness or uncertainty; the agent cannot be sure of the outcome.
 - *Example: Weather Prediction or Dice Games.*

3. History Dependency

- **Episodic:** The agent's experience is divided into atomic "episodes." The action taken in one episode does not affect the next.
 - *Example: Image Classification* (identifying one image doesn't affect the next).
- **Sequential:** Current decisions have long-term consequences. The agent must think ahead.

- *Example:* **Driving a Car** or **Chess**.

4. Stability

- **Static:** The environment does not change while the agent is deliberating or "thinking."
 - *Example:* **Crossword Puzzles**.
- **Dynamic:** The environment changes continuously while the agent is deciding on an action.
 - *Example:* **Air Traffic Control** or **Stock Market**.

5. State Space

- **Discrete:** There are a limited, distinct number of states and actions.
 - *Example:* **Tic-tac-toe** (fixed number of squares).
- **Continuous:** The environment has infinite possibilities, often involving real-valued numbers (time, distance, speed).
 - *Example:* **Robot Navigation** (precise angles and velocity).

6. Agent Cardinality

- **Single-Agent:** Only one agent is acting in the environment.
 - *Example:* **Solitaire** or a **Thermostat**.
- **Multi-Agent:** Multiple agents interact, either cooperatively or competitively.
 - *Example:* **Online Auction Systems** or **Football**.

7. Rule Awareness

- **Known:** The agent knows the "laws of physics" or the rules of the environment.
 - *Example:* **Video Games** (pre-programmed rules).
- **Unknown:** The agent must explore and learn how the environment works.
 - *Example:* **New Planet Exploration**.

Comparative Summary Table

Property	Type A	Type B	Complexity Level
Observability	Fully Observable	Partially Observable	Partially is Harder
Nature	Deterministic	Stochastic	Stochastic is Harder
Task Type	Episodic	Sequential	Sequential is Harder
Change	Static	Dynamic	Dynamic is Harder
State Type	Discrete	Continuous	Continuous is Harder
Agents	Single-Agent	Multi-Agent	Multi-Agent is Harder
Knowledge	Known	Unknown	Unknown is Harder

💡 **Exam Strategy:** When writing this in an exam, use the table above as your conclusion. It demonstrates a high-level understanding of how these properties affect the difficulty of building the AI.